

EARLY-STAGE RESULTS

# Emotion-Mediated Personalized Image Aesthetic Assessment

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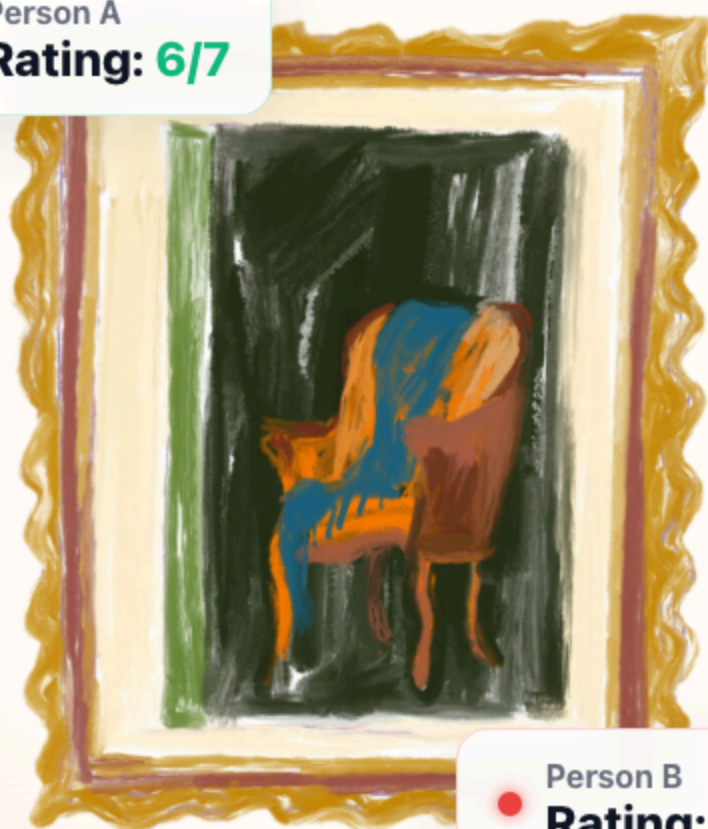
Predicting individual aesthetic preferences through emotional responses



# The Core Problem.



Person A  
Rating: 6/7



Person B  
Rating: 2/7



The exact same image receives completely different beauty scores from different people.



## Subjectivity of Beauty

Beauty is in the eye of the beholder. If we show the same picture to different observers, their responses exhibit high variance. Predicting a single average rating fails to capture individual preference.

## What is PIAA?

**Personalized Image Aesthetic Assessment (PIAA)** is a specialized field in AI. Instead of predicting a generic general aesthetic score, the goal is to train an AI model capable of predicting the specific aesthetic score that a **particular individual** would assign to an image.

Our Ultimate Goal:

**Predict the score for one specific person**





# Our Proposed Core Idea.

## TRADITIONAL METHODS

BLACK BOX

### Direct Prediction (Black Box)

Most previous methods map the image directly to a personal aesthetic score, operating as an uninterpretable black box.

Image → AI → Score

## OUR APPROACH

EXPLAINABLE

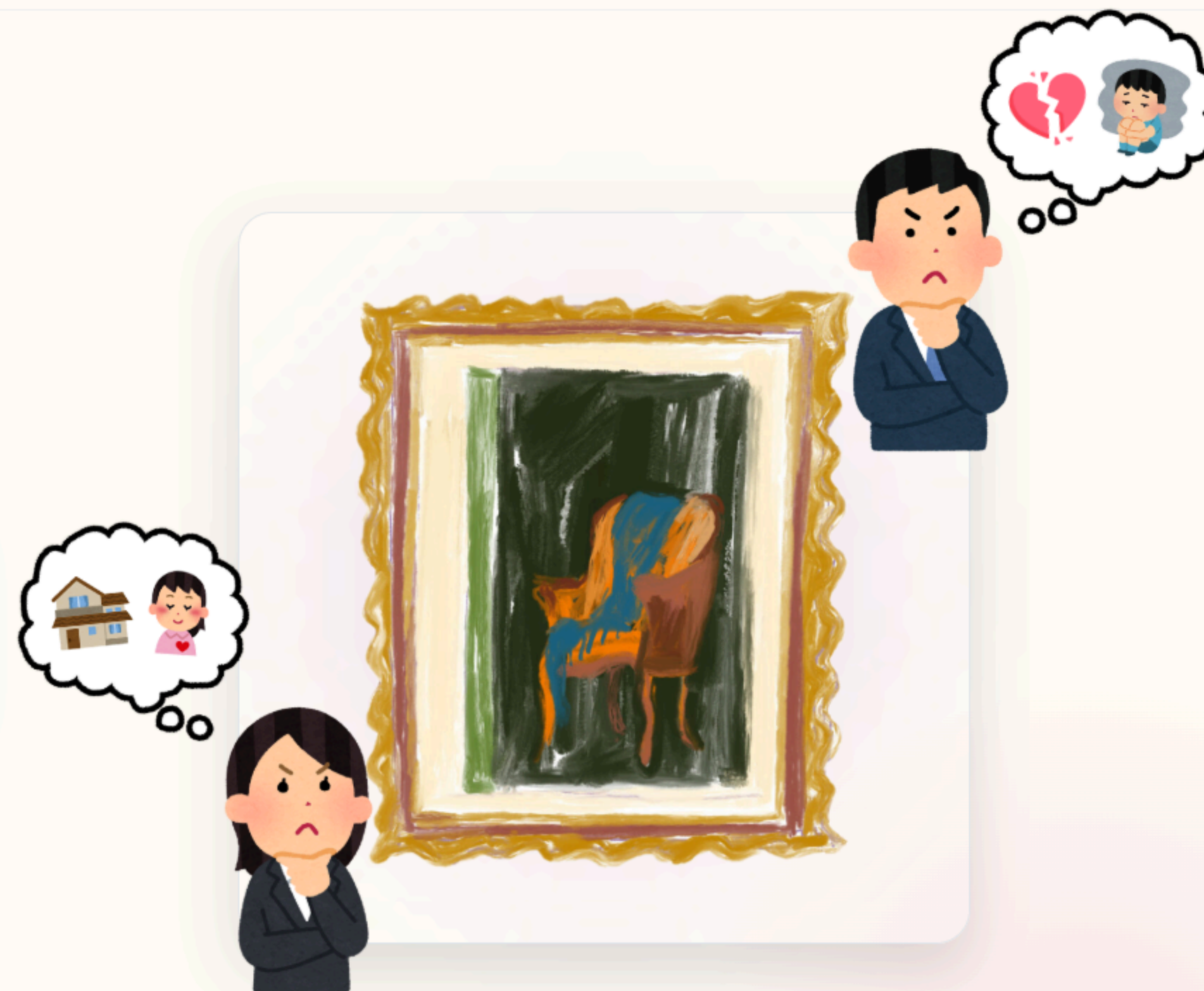
### Emotion-Mediated Prediction

We predict emotions as intermediate steps. Each user is modeled by a **personal formula** (emotion weights).

Image → Emo → formula → Score

## Grounded in Psychology Research

Based on human pathways: people judge beauty through emotional responses (e.g. nostalgic, amused, impressed), not directly from pixels.



# The Prediction Pipeline.



## Input Image

Any artwork, fashion photograph, or landscape scene.



TRAINED ON ALL USERS

## Step 1: Shared AI

A generic model that predicts the 9 emotional response levels for the image (e.g. nostalgia = 4/5).



9 WEIGHTS PER USER

## Step 2: Personal Formula

Each person has a unique set of weights representing how much they care about each emotion when judging beauty.

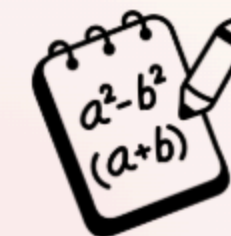


## Personal Score

A custom aesthetic prediction tailored to that specific user.



**Technical Note:** Step 2 is not a trained deep model — it is a simple **Ridge Regression formula** (9 weights per user) computed almost instantly to map emotions to the personal beauty score.



# How This Works in Practice.

| STAGE                                                                 | WHAT'S NEEDED                                      | WHERE IT COMES FROM                                                                         |
|-----------------------------------------------------------------------|----------------------------------------------------|---------------------------------------------------------------------------------------------|
| <b>Stage 1: Train Model 1</b><br>(One-time setup by researchers)      | Images + emotional ratings from a large population | <b>XPASS-Vis Dataset</b><br>(87,836 ratings already available; no custom collection needed) |
| <b>Stage 2: New User Signs Up</b><br>(Personalizing the formula)      | ~50 to 100 beauty ratings from that specific user  | <b>Direct User Annotation</b><br>(User rates a small set of onboarding images)              |
| <b>Stage 3: Predict on New Image</b><br>(Instant personal assessment) | Just the query image                               | —                                                                                           |

# The XPASS-Vis Dataset.

## IMAGES

6,526

Art, Fashion, Landscape



## PARTICIPANTS

129

Individual evaluators

## TOTAL RATINGS

87,836

Fully labeled examples

## LABELS PER RATING

1 + 9

1 Beauty (1-7) + 9 Emotions (1-5)



## Observer Metadata Collected

- **Demographics**  
Nationality, Educational background
- **Personality Traits (TIPI)**  
Big Five: Openness, Conscientiousness, Extraversion, Agreeableness, Emotional Stability
- **AESTHEMOS Emotions**  
9 Aesthetic emotional responses (categorized into Positive, Cognitive, and Negative groups)





# What is AESTHEMOS?

## Aesthetic Emotion Scale

**AESTHEMOS** (Schindler et al., 2017) is a validated psychology questionnaire designed to measure the specific emotional responses people feel when experiencing beauty and art.

Rather than random descriptive tags, XPASS-Vis selected **9 core dimensions** that represent three primary aesthetic emotion families. This provides solid scientific grounding for our explainable pipeline.

## The 3 Emotion Families

### ● POSITIVE EMOTIONS

Liked it

Found it beautiful

Impressed

Motivated

Amused

### ● COGNITIVE EMOTIONS

Nostalgic

Intellectually challenged

### ● NEGATIVE EMOTIONS

Sad

Distasteful

# What is TIPI / Big Five?

**TIPI (Ten-Item Personality Inventory)** is a brief, scientifically validated questionnaire developed by **Gosling et al. (2003)**. It is widely used in psychology to measure the **Big Five personality traits**. XPASS-Vis collected these traits for all 129 participants, allowing us to evaluate if individual aesthetic formulas correlate with real-world personality (RQ3).

TRAIT 1

## Openness

Curiosity, openness to new experiences, and artistic appreciation

TRAIT 2

## Conscientiousness

Organization, self-discipline, and goal-directed behavior

TRAIT 3

## Extraversion

Sociability, assertiveness, energy, and talkativeness

TRAIT 4

## Agreeableness

Altruism, kindness, trust, and cooperative spirit

TRAIT 5

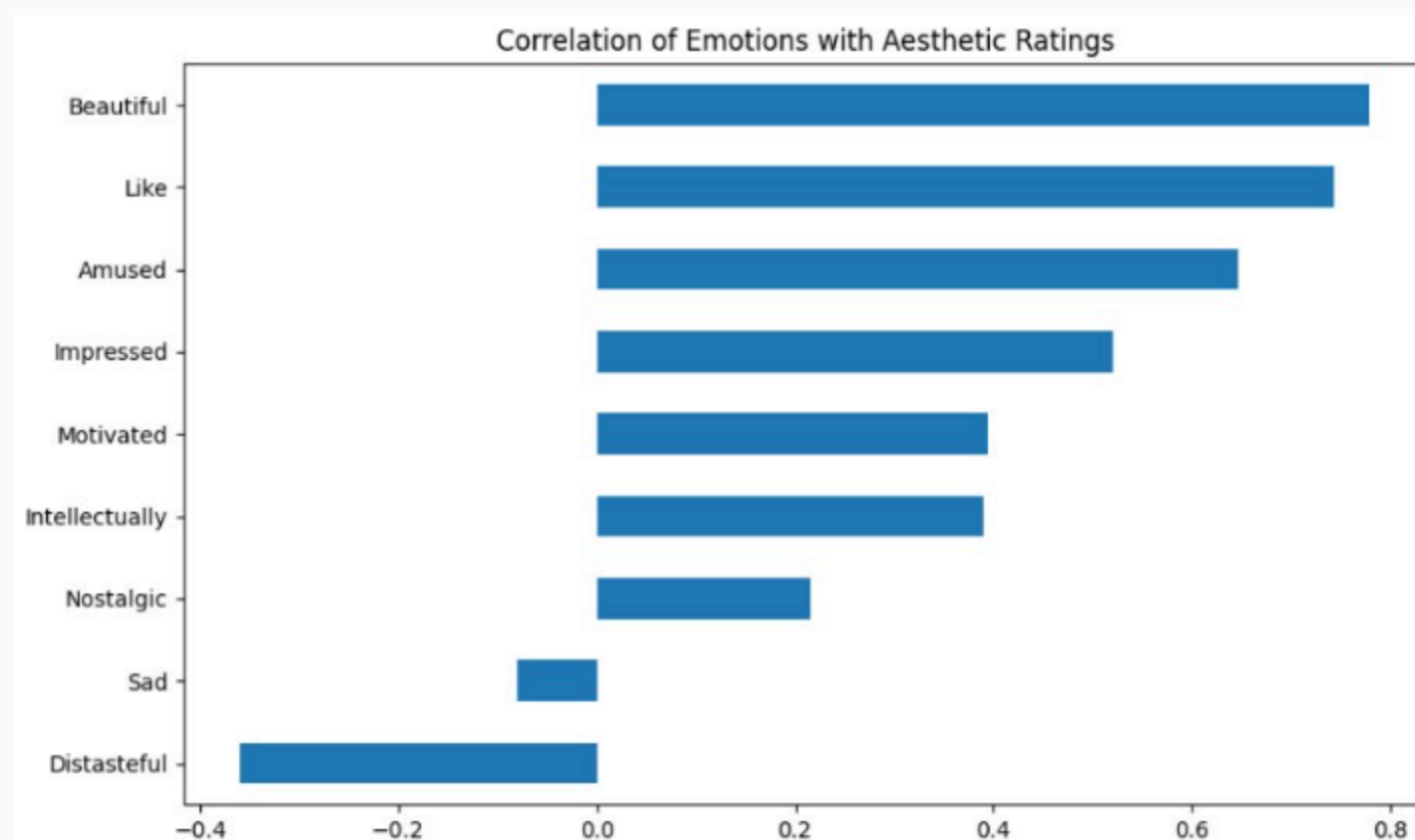
## Emotional Stability

Commonly reversed as Neuroticism — calm vs. tendency to experience anxiety



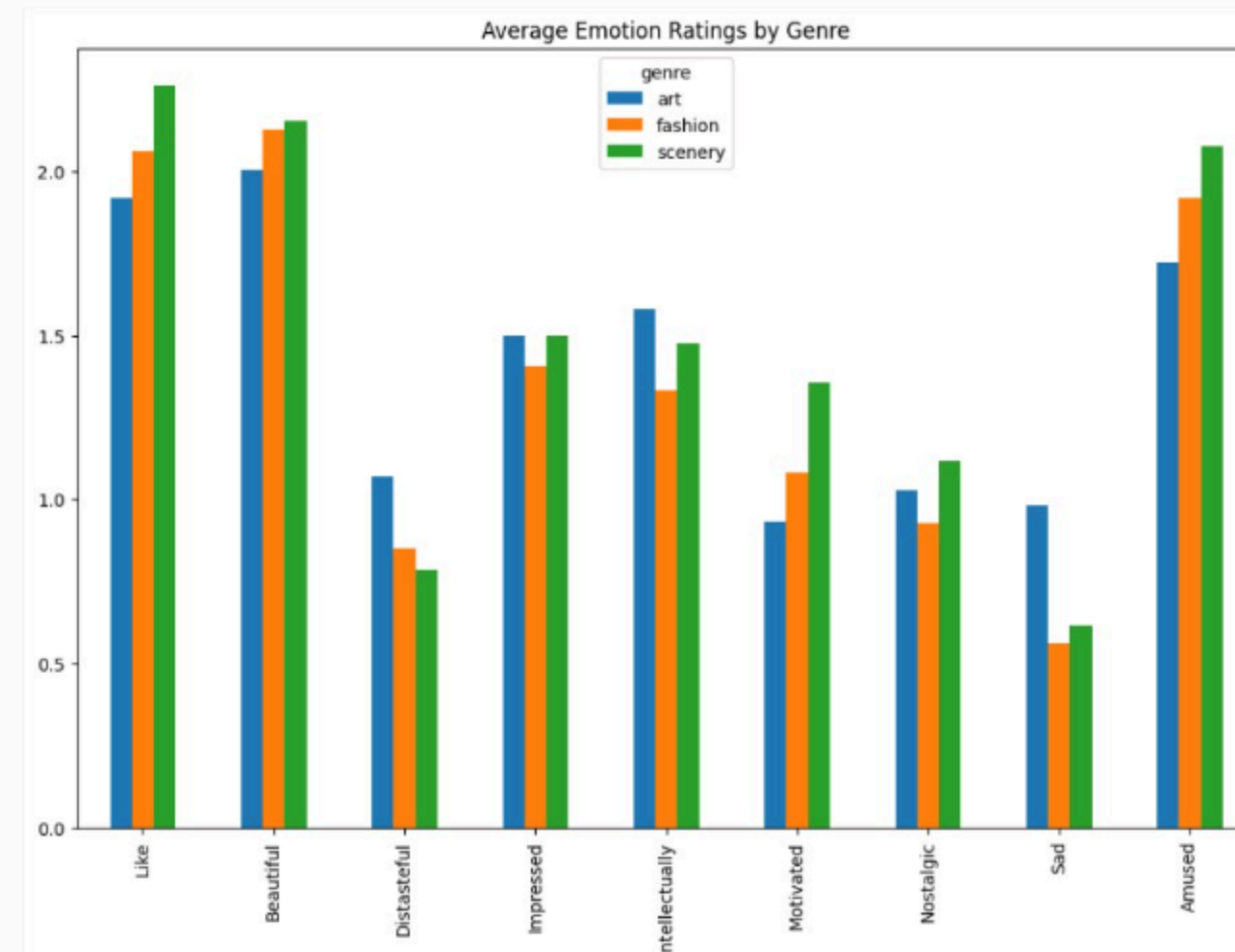
# Dataset Visualizations.

## Emotion Correlations with Aesthetics



Positive emotions like 'Beautiful' and 'Like' show high correlation with overall aesthetics, while negative emotions like 'Distasteful' are highly negatively correlated.

## Average Emotion Ratings by Genre



Aesthetic ratings vary significantly across different genres (Art, Fashion, Scenery), with positive emotions scoring higher on average for Scenery images.

# Research Questions.

## Q1 Accuracy vs. Direct Prediction

Can emotion-mediated modeling perform as accurately as direct aesthetic prediction?

Comparing our explainable two-stage pipeline against traditional black-box deep models to see if we sacrifice performance for transparency.



## Q2 Mediation Redundancy

Should 'Liked' or 'Found it beautiful' be excluded as mediating emotions?

Two emotions ('Liked it', 'Found it beautiful') are very close to the final score itself. Should we keep them, or remove them?



## Q3 Psychological Grounding

Does the personal formula correlate with real-world personality traits?

Testing if users' personal formulas reflect real personality traits (e.g. Openness to Experience correlating with 'Intellectually Challenged' weight).

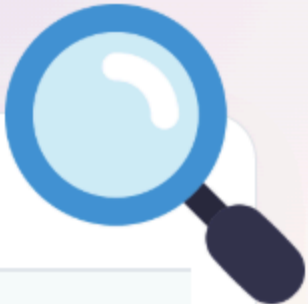
## Q4 Cross-Category Consistency

Is a person's 'personal emotion formula' consistent across different image categories?

If a user appreciates fine art due to nostalgia, do they use the same criteria for fashion or landscape photos, or does their formula shift?



# Experiment 1: Is the Idea Even Possible?



| METHOD / DATASET INPUT                                                               | ART   | FASHION | LANDSCAPE | AVERAGE |
|--------------------------------------------------------------------------------------|-------|---------|-----------|---------|
| <b>All 9 emotions</b><br>Using the full set of emotional responses                   | 0.819 | 0.838   | 0.843     | 0.833   |
| <b>7 emotions (Real emotions)</b><br>Removed circular proxies: 'liked' & 'beautiful' | 0.716 | 0.705   | 0.740     | 0.720   |
| <b>Best previous method (ICI)</b><br>Shi et al., 2024 (without emotions)             | 0.493 | 0.319   | 0.458     | 0.423   |

\* Values represent Concordance Correlation Coefficient (CCC) score, max = 1.0 (higher is better).

Theoretical Ceiling: 0.720

## ✓ Key Experimental Insight

Even after removing circular proxies (reducing from 9 emotions to 7 "pure" aesthetic emotions), our pipeline still achieves a high prediction accuracy of **0.720 CCC**. This is extremely close to the theoretical ceiling of **0.720** and nearly doubles the performance of the best previous method, ICI (**0.423**), proving the feasibility of emotion-mediated personalized assessment.

# Experiment 2: Where does Personalization Come From?



## GAP 1 — FEELING DIFFERENTLY (0.39)

Different people look at the exact same image and feel completely different emotions (nostalgic vs neutral). This accounts for **88%** of the personalization drop.

## GAP 2 — CARING DIFFERENTLY (0.05)

Even when feeling identical emotions, observers value/weight them slightly differently when judging beauty. This accounts for only **12%**.

**Crucial Insight:** Taste personalization comes mainly from how people *feel* about an image, not how they *weigh* those feelings.

\* ICI baseline performance (0.423) is referenced from Table 5 of the XPASS-Vis paper.



# Experiment 3a: Comparing the 4 Approaches.

| APPROACH (SUPPORT = 100)                                                                                         | ART   | FASHION | LANDSCAPE | AVERAGE      |
|------------------------------------------------------------------------------------------------------------------|-------|---------|-----------|--------------|
| <b>Direct (No Emotion)</b><br>Baseline model predicting beauty score straight from CLIP visual features          | 0.379 | 0.191   | 0.312     | <b>0.294</b> |
| <b>Population-emo</b><br>Predicts general average emotions first, then maps via personal formula                 | 0.387 | 0.245   | 0.373     | <b>0.335</b> |
| <b>Personal-emo</b> OVERFIT<br>Attempts to predict personalized emotions using only user's 100 images (overfits) | 0.337 | 0.118   | 0.254     | <b>0.237</b> |
| <b>Hybrid (Proposed)</b> BEST<br>Starts with stable average emotions and adds a learned personal correction      | 0.420 | 0.234   | 0.347     | <b>0.334</b> |

\* Values represent Concordance Correlation Coefficient (CCC) score, higher is better.

ICI (State-of-the-Art): 0.423   Theoretical Ceiling (P-oracle): 0.720

# Experiment 3b: Improvement Summary.

| MODEL METHOD                                                                    | SUPPORT = 10 | SUPPORT = 25 | SUPPORT = 50 | SUPPORT = 100 |
|---------------------------------------------------------------------------------|--------------|--------------|--------------|---------------|
| <b>Direct (No Emotion)</b><br>Predicting aesthetic score directly from pixels   | 0.109        | 0.187        | 0.245        | 0.294         |
| <b>Through Emotion + Correction (Hybrid)</b><br>Proposed emotion-mediated route | 0.153        | 0.238        | 0.299        | 0.334         |
| <b>Improvement</b><br>Relative performance gain                                 | +40%         | +27%         | +22%         | +14%          |

\* Support represents the number of images rated by the person used to train their personalized formula.

ICI State-of-the-Art: 0.423    Theoretical Ceiling: 0.720

### Highest Gain in Scarce Data

The improvement is largest (+40%) when we have the least data (only 10 images per user). This highlights how using emotional mediation acts as a strong prior, preventing overfitting.

### Consistent Outperformance

Going through predicted emotions and applying correction consistently beats direct score prediction at all support levels (10 to 100 images).

### Room for Improvement

We are currently at 0.334, which is below ICI (0.423). However, the theoretical ceiling is 0.720, indicating that refining the emotion predictor offers a huge path forward.

# Research Challenges.



## Predicting Personal Emotions

An image can make different people feel very different things. A general AI model typically only learns the average human reaction, not individual emotional subjectivity.



## Rare Emotion Sparsity

Certain emotions occur very rarely in particular domains, causing data imbalance that is hard to train.



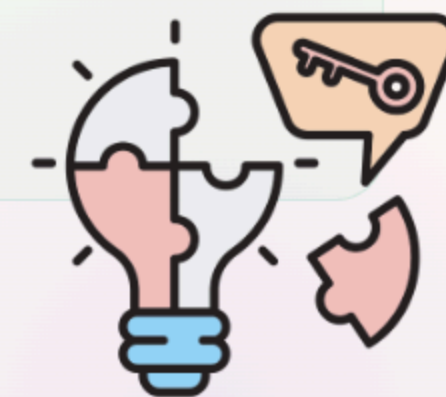
## Limited Personalized Data

Each participant rated ~100 images. Fitting the 9-dimensional personal formula on such limited samples poses a significant risk of overfitting.



## Evaluation Alignment

To compare fairly against existing baselines, we need identical cross-validation data splits across all models.



# Next Research Steps.

01

## 5-Fold Results Complete ✓

We've completed the full 5-fold evaluation across all 129 users — results shown today are final, not preliminary.

02

## Correlate with Personality (Big Five)

Cross-reference each person's emotion weightings (personal formula) with their Big Five personality profiles already present in XPASS-Vis.

03

## Test Domain Consistency

Analyze whether an individual's personal formula stays consistent when evaluating different domains (e.g. switching from landscape to fashion).

04

## Paper Write-up & Submission

Synthesize findings, document the explainability advantages of emotion mediation, and start drafting the paper.





# Questions for Discussion.

01

## Improving Personal Emotion Prediction

In Experiment 3, we found that the bottleneck is predicting personalized emotions accurately from pixels. Do you think we should focus on refining this (e.g., trying different image features or more complex models), or is there another direction you suggest first?

02

## Access to ICI Baseline Code

Would it be possible to obtain the code for the ICI baseline (Shi et al., 2024)? Currently we are comparing against the results reported in their paper. Accessing the code would allow for a direct, fair evaluation on our exact cross-validation splits.

# Thank You!

Happy to take any questions or suggestions.

# Literature & Related Work.



| PAPER                                        | CONTRIBUTION                                                                                                      | RELATION TO OUR WORK                                                                                                  |
|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| ICI (Shi et al., 2024)                       | Predicts personal aesthetic score directly using image features and user metadata.                                | Ours uses emotions as an explainable intermediate step rather than a black-box end-to-end mapping.                    |
| HNEF (Lan et al., 2024)                      | Leverages emotional attributes as supplementary features to improve general (non-personal) aesthetic predictions. | Ours focuses on personalization, using emotional reactions as the primary signal to capture individual preference.    |
| Iigaya et al. (2021, Nature Human Behaviour) | Demonstrated that human aesthetic preferences can be modeled as a weighted linear combination of visual features. | Serves as the theoretical backing for our 'personal formula' concept (modeling beauty as a weighted sum of emotions). |
| Schindler et al. (2017)                      | Developed AESTHEMOS, the aesthetic emotion scale our 9 emotions are based on.                                     | Provides the psychological grounding for which emotions we use in our design.                                         |
| Liu & Wagemans (2026, arXiv:2603.18108)      | Interpretable IAA via concept activation vectors + residual predictor.                                            | Our 'Hybrid' residual-correction idea follows a similar interpretable-core + residual structure.                      |
| Ryu & Yanaka (2026, arXiv:2604.11374)        | Shows VLM hidden representations work well for PIAA via simple linear probes.                                     | A candidate stronger image feature for our emotion predictor (Phase 1B direction).                                    |

# Evaluation Metrics (CCC/SCC).

## CCC — Concordance Correlation Coefficient

$$\text{CCC} = 2 \cdot \text{cov}(y, \hat{y}) / ( \text{var}(y) + \text{var}(\hat{y}) + (\text{mean}(y) - \text{mean}(\hat{y}))^2 )$$

checks ranking AND absolute scale — penalizes if predictions are shifted or scaled differently from the true scores

## SCC — Spearman rank correlation

SCC = Pearson correlation of  $\text{rank}(y)$  and  $\text{rank}(\hat{y})$

checks ranking only — did we put the images the user would rate highest near the top, regardless of exact scores?

## Concordance Correlation Coefficient

Measures both **scale alignment** and **ranking alignment**. Unlike standard correlation, it penalizes predictions if they are shifted (e.g. consistently too high) or scaled differently.

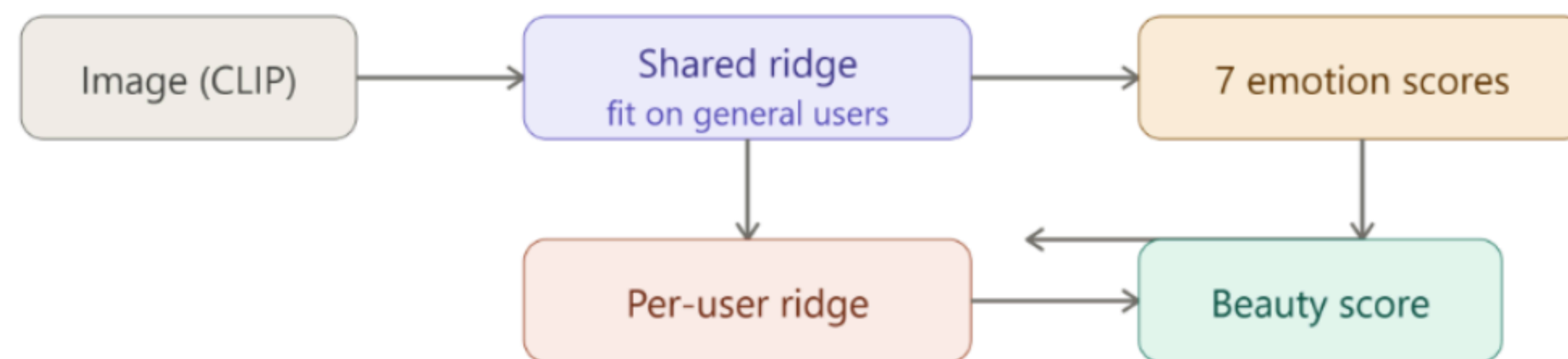
## Spearman Rank Correlation

Measures **ranking order only**. It ignores the exact numeric ratings and only checks if the predicted order matches the observer's relative preferences.

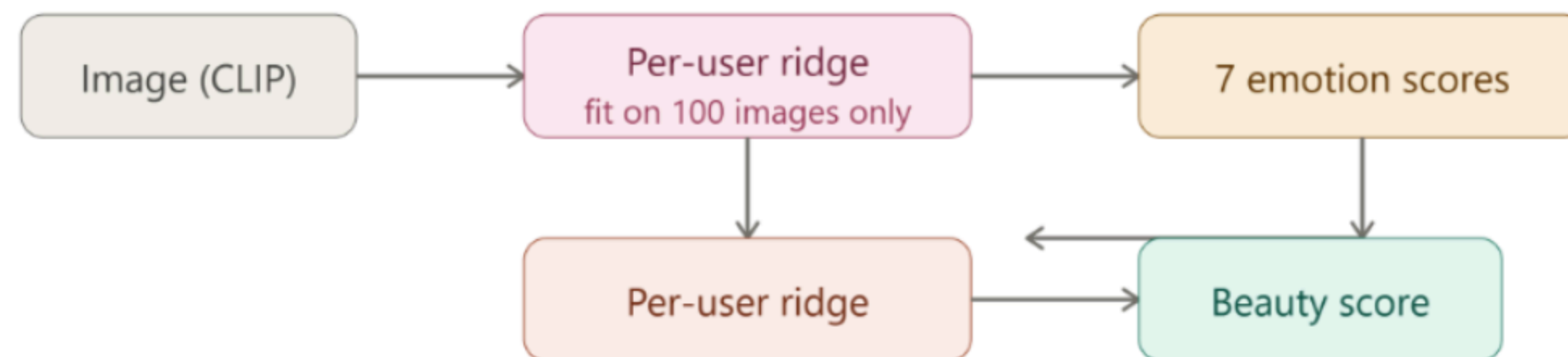


# Experiment 3: 4 Approaches Detail.

## 2. Population-emo



## 3. Personal-emo



## 4. Hybrid (best)

### Approach Comparison Highlights

**1. Direct:** Baseline, maps visual embeddings directly to personal scores via per-user ridge.

**2. Population-emo:** Shared emotion prediction module. Personalization occurs at the weight mapping phase.

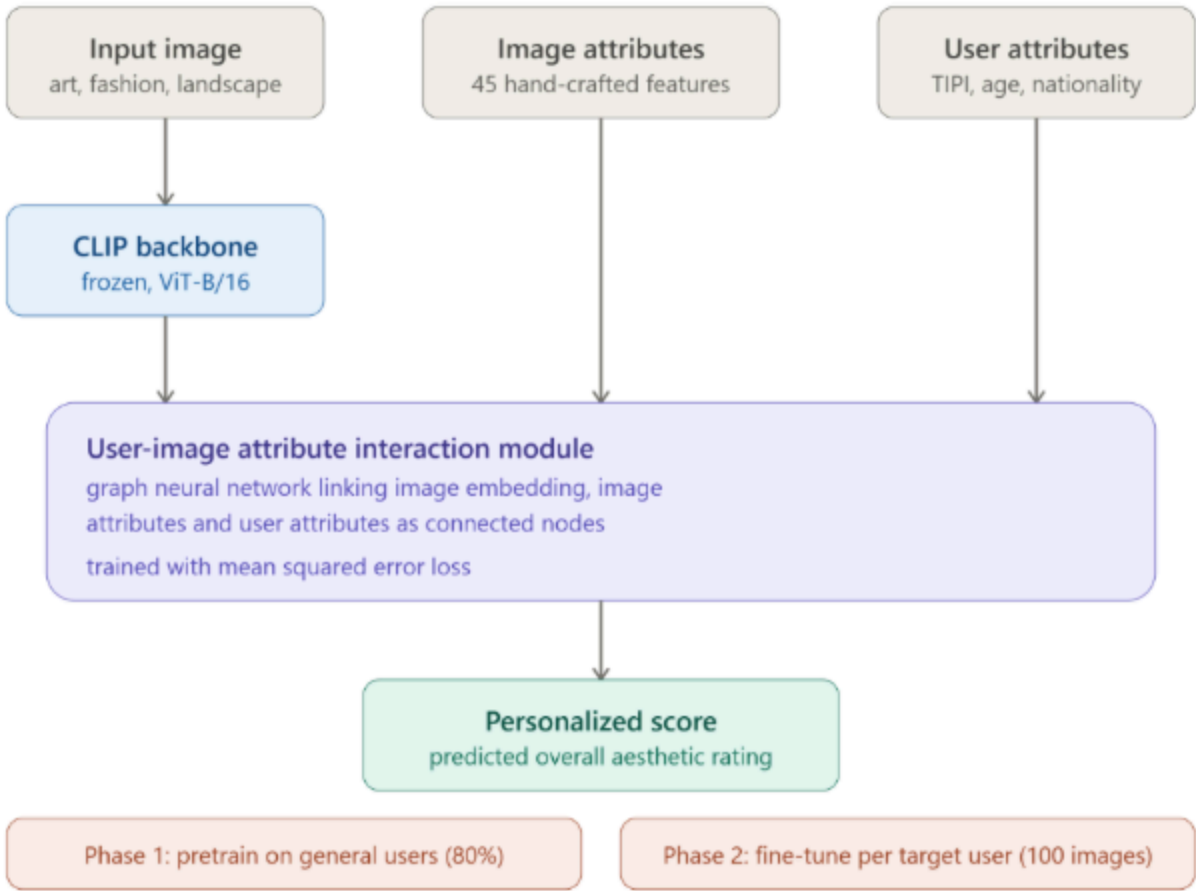
**3. Personal-emo:** Tries to fit both models per-user. Underperforms due to data scarcity (~100 images).

**4. Hybrid (Best):** Combines stable shared population predictions with small per-user corrections.

# Models & Hyperparameters.

| Component                   | Method                  | Key Settings / Details                                                                           |
|-----------------------------|-------------------------|--------------------------------------------------------------------------------------------------|
| Image Features              | CLIP ViT-B/16           | Frozen, pretrained by OpenAI — produces 512-dimensional embedding per image                      |
| Emotion Prediction (Step 1) | Ridge Regression        | Regularization strength alpha selected via cross-validation from 13 values between 0.01 and 1000 |
| Personal Formula (Step 2)   | Ridge Regression        | Same cross-validation approach for individual user emotion weighting                             |
| Evaluation Setup            | 5-Fold User-Level Split | 80% users for training, 20% held out as new users — repeated 5 times so every user is tested     |
| Performance Metrics         | CCC & SCC               | CCC checks agreement in both ranking & scale; SCC checks ordinal ranking agreement only          |

# Baseline: The ICI Architecture.



ICI model flow: graph neural network mapping image embeddings, visual features, and user metrics.

## ICI vs. Our Proposed Approach

| Feature         | ICI Baseline                                                  | Our Approach                                   |
|-----------------|---------------------------------------------------------------|------------------------------------------------|
| Input           | Image + image attributes (45-dim) + user attributes (114-dim) | Image only                                     |
| Core Module     | Graph Neural Network                                          | CLIP + Ridge Regression                        |
| Personalization | Fine-tune entire GNN network per user (needs 100 images)      | Fit a simple 9-number formula per user         |
| Output          | Score (black box)                                             | Score + readable emotion weights (explainable) |